GA4 + Google Tag Manager Tracking Setup

Tracking plan and QA report · one GTM container, one GA4 property, two sites · Prepared by Ashwin Kumar

Events tracked	Tags	Triggers	Variables	QA result
11 custom + GA4 enhanced measurement	7	6	3 custom + 6 built-in	11 of 11 custom events confirmed on the live sites

1 · Summary

• A single Tag Manager container sends engagement events from a live portfolio site and the full e-commerce funnel from a demo store into one GA4 property.
• Every rule lives in Tag Manager. The sites carry only the container snippet; the demo store adds standard GA4 e-commerce dataLayer pushes.
• All 11 custom events were triggered on the published sites and checked in the browser, and GA4 Realtime showed data arriving with purchase counted as a key event.
• QA caught one defect: the scroll trigger was imported without its thresholds and never fired. It was fixed and republished as container version 3.
• No personal data is collected: the sign-up form sends only an event name and a form name.

2 · Architecture

Page pushes an event to the dataLayer, or the visitor clicks a link or scrolls → a **Tag Manager trigger** matches it → a **GA4 event tag** forwards it with its parameters → **GA4** stores it and reports on it. The demo store is a mock shop with invented products; nothing is sold and no payment is taken.

3 · Tracking plan

Event	Site	Fires when	Parameters	Business question it answers
page_view	Both	Every page load	page_location, page_title	How many people visit, and from where?
project_click	Portfolio	Click on a GitHub project link	link_url, link_text	Which projects get attention?
dashboard_click	Portfolio	Click on a Looker Studio link	link_url, link_text	Do visitors open the live dashboards?
contact_click	Portfolio	Click on email, phone or LinkedIn	contact_method, link_url	How many visitors try to get in touch, and how?
scroll_depth	Both	Page scrolled 25 / 50 / 75%	percent_scrolled	How far down the page do people read?
view_item_list	Store	Product grid shown	items[]	Which products are seen?
view_item	Store	Product detail opened	currency, value, items[]	Which products draw interest?
add_to_cart	Store	Add to cart clicked	currency, value, items[]	Where does buying intent start?
begin_checkout	Store	Checkout clicked	currency, value, items[]	How many carts reach checkout?
purchase	Store	Test order placed	transaction_id, currency, value, items[]	Revenue, orders and ROAS
generate_lead	Store	Sign-up form submitted	form_name	How many leads does the page produce?

GA4 Enhanced Measurement additionally records 90% scroll, outbound clicks, form starts and file downloads.

4 · Container contents

Type	Name	Purpose
Tag	Google Tag - GA4 Config	Loads GA4 on every page (Initialization - All Pages)
Tag	GA4 Event - project_click / dashboard_click / contact_click	One event tag per link-click trigger
Tag	GA4 Event - scroll_depth	Sends the scroll threshold reached
Tag	GA4 Event - Ecommerce (dataLayer)	One tag for all five e-commerce events; event name and items read from the dataLayer
Tag	GA4 Event - generate_lead	Lead form submission
Trigger	Link Click x3	GitHub project links; Looker Studio links; mailto, tel and LinkedIn links
Trigger	Scroll Depth - 25, 50, 75	Vertical scroll thresholds
Trigger	Custom Event x2	E-commerce event names (regex); generate_lead
Variable	Const - GA4 Measurement ID	Measurement ID kept in one place
Variable	JS - Contact Method	Turns the clicked link into email / phone / linkedin
Variable	DLV - form_name	Reads the form name from the dataLayer

5 · QA results

Tested on the published sites on 6 October 2026. "Network" means a request to google-analytics.com/g/collect carrying that event name was observed in the browser.

Event	Test action	Result	Evidence
page_view	Loaded both sites	Pass	Network; GA4 Realtime (4 views across both page titles)
view_item_list	Loaded the store	Pass	Network, with items; GA4 Realtime
view_item	Opened Forest Stripe Tee	Pass	Network (value 999)
add_to_cart	Added two products	Pass	GA4 Realtime (count 2)
begin_checkout	Clicked Checkout	Pass	GA4 Realtime (count 1)
purchase	Placed test order, value 1,998 INR	Pass	Network (value 1998, items); GA4 Realtime key event
generate_lead	Submitted sign-up form	Pass	Network (form_name = launch_updates)
contact_click	Clicked email and LinkedIn links	Pass	Network (contact_method = email, linkedin); GA4 Realtime (count 2)
project_click	Clicked a GitHub project link	Pass	Network (link_url = github.com project)
dashboard_click	Clicked a Looker Studio link	Pass	Network (link_url = Looker Studio report)
scroll_depth	Scrolled the portfolio past 50%	Pass after fix	Failed on version 2 (no thresholds). On version 3 the trigger fired at 25% and 50% and the hit was sent in a batched request

6 · Recommended next steps

• Register contact_method, percent_scrolled and form_name as custom dimensions in GA4 so they appear in standard reports.
• Mark generate_lead and contact_click as key events alongside purchase.
• Build a funnel exploration: view_item_list → view_item → add_to_cart → begin_checkout → purchase.
• Add a consent banner and Consent Mode before using the setup on a site with visitors from regions that require consent.
• Link the GA4 property to Google Ads and import purchase and generate_lead as conversions.